Synopsis

Project Title: Heart Disease Prediction System Using Machine Learning

Name: Aditya Maurya

University Roll No: 2028624

Sec - Q

Name: Sarthak Pandey

University Roll No: 2028487

Sec - Q

Name: Piyush Shukla

University Roll No: 2028465

Sec - Q

1. Introduction

Heart disease is one of the most dangerous health problems affecting millions of people worldwide. Early prediction and diagnosis of heart disease can reduce medical risks and improve patient survival rates. Machine Learning techniques are becoming highly useful in healthcare because they can analyze medical data quickly and identify patterns related to diseases. This project focuses on developing a Heart Disease Prediction System using Machine Learning algorithms. The system predicts whether a patient is likely to have heart disease based on medical and lifestyle-related features such as age, blood pressure, cholesterol, glucose level, BMI, smoking habits, and family history. The project demonstrates the complete workflow of a Machine Learning system including dataset creation, data preprocessing, visualization, model training, testing, and prediction.

2. Problem Statement

The objective of this project is to design a machine learning-based system capable of predicting heart disease using patient clinical data. The system should: • Create and manage a heart disease dataset. • Handle missing values and noisy data. • Detect and remove outliers. • Visualize patterns in the dataset. • Train machine learning models. • Predict whether a patient has heart disease or not. • Evaluate the performance of models using different metrics.

3. Objectives

The major objectives of this project are: • To generate a dataset with multiple medical features. • To perform data cleaning and preprocessing. • To understand Exploratory Data Analysis (EDA). • To implement Logistic Regression and KNN algorithms. • To compare model performance. • To perform prediction using new patient data. • To understand practical implementation of Machine Learning.

4. Scope of Project

The Heart Disease Prediction System can be useful in healthcare organizations, hospitals, and medical research systems. The project helps in: • Early disease prediction. • Medical risk analysis. • Clinical decision support. • Reducing manual analysis. • Assisting doctors with patient evaluation. The project demonstrates how Artificial Intelligence and Machine Learning can improve healthcare systems.

5. Technologies Used

Python: Python is the main programming language used in this project because it provides simple syntax and powerful libraries for Machine Learning and Data Analysis. Pandas: Pandas is used for dataframe handling, missing value handling, cleaning operations, and preprocessing. NumPy: NumPy is used for numerical computations and generation of random datasets. Matplotlib: Matplotlib is used to create histograms, scatter plots, comparison graphs, and visualization charts. Seaborn: Seaborn is used for advanced statistical graphs such as heatmaps, pairplots, and boxplots. Scikit-learn: Scikit-learn is used for feature scaling, train-test split, Logistic Regression, KNN, model evaluation, and prediction.

6. System Requirements

Hardware Requirements: • Minimum 4 GB RAM • Intel i3 Processor or higher • 500 MB storage space Software Requirements: • Python 3.x • Jupyter Notebook / VS Code • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn

7. Methodology

The project follows a step-by-step methodology: 1. Dataset Generation 2. Dataset Storage in CSV File 3. Missing Value Handling 4. Duplicate Removal 5. Outlier Detection 6. Outlier Removal 7. Feature Scaling 8. Data Visualization 9. Feature Selection 10. Train-Test Split 11. Model Training 12. Model Testing 13. Prediction Each step improves the quality and performance of the Machine Learning system.

8. Dataset Description

A synthetic heart disease dataset containing 2000 rows and 25 features was created. The dataset includes medical and lifestyle-related attributes. The target column is HeartDisease: • 1 → Patient has heart disease • 0 → Patient does not have heart disease The target column was generated using medically related conditions such as high BP, high cholesterol, high BMI, smoking habits, and glucose level.

9. Dataset Features

The dataset contains the following important features: • Age • Sex • Chest Pain Type • Resting Blood Pressure • Cholesterol • Fasting Blood Sugar • ECG Result • Maximum Heart Rate • Exercise Angina • BMI • Smoking Habit • Alcohol Consumption • Stress Level • Sleep Hours • Family History • Glucose Level • Oxygen Level • Heart Rate These features play an important role in predicting heart disease risk.

10. Data Cleaning and Preprocessing

Data preprocessing is one of the most important stages in Machine Learning. Raw datasets usually contain missing values, duplicate records, and noisy data. In this project: • Missing values were intentionally added. • Missing values were handled using different methods. • Duplicate records were removed. • Outliers were detected and removed. • Feature scaling was performed. Proper preprocessing improves model accuracy and reliability.

11. Missing Value Handling

Missing values are common in real-world datasets. The following techniques were used: Mean Imputation: Missing values were replaced using the average value of the column. Median Imputation: Missing values were replaced using the median value. Mode Imputation: Most frequent values were used to fill missing entries. Row Removal: Rows containing excessive missing values were removed.

12. Duplicate Removal

Duplicate records reduce the quality of datasets and may negatively affect prediction performance. Duplicate rows were detected using Pandas functions and removed to maintain data quality and improve the accuracy of machine learning models.

13. Outlier Detection and Removal

Outliers are extreme values that differ significantly from other observations. Outliers can reduce model performance. The Interquartile Range (IQR) method was used: • Q1 and Q3 values were calculated. • IQR was computed. • Values outside the acceptable range were identified as outliers. Outliers were removed to improve data consistency.

14. Feature Scaling

Feature scaling ensures that all numerical features are within a similar range. Two scaling techniques were used: StandardScaler: Transforms data into standard normal distribution. MinMaxScaler: Converts values into a range between 0 and 1. Scaling is important for algorithms such as KNN because distance calculations depend on feature magnitude.

15. Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps understand dataset characteristics. The following visualizations were performed: • Histograms • Scatter plots • Boxplots • Pairplots • Correlation heatmaps EDA helped identify feature relationships, distributions, and outliers.

16. Histograms and Distribution Analysis

Histograms were used to understand the distribution of features such as: • Age • Cholesterol • Blood Pressure • Glucose The graphs showed how values were distributed across the dataset and helped identify skewness and spread.

17. Scatter Plot Analysis

Scatter plots were used to analyze relationships between variables. Example: Age vs Cholesterol Scatter plots helped visualize whether an increase in one feature affects another feature.

18. Heatmap and Correlation Analysis

Correlation heatmaps were generated using Seaborn. The heatmap displayed relationships among all features. Strong correlations were observed between: • Cholesterol and HeartDisease • Blood Pressure and HeartDisease • BMI and HeartDisease Heatmaps helped in feature selection and understanding feature importance.

19. Feature Selection

Feature selection helps improve model performance by selecting important attributes. Features such as: • Age • Cholesterol • RestingBP • BMI • Smoking • Glucose showed strong impact on heart disease prediction. Unnecessary features can reduce performance and increase computational complexity.

20. Train-Test Split

The dataset was divided into training and testing data. • 80% data was used for training. • 20% data was used for testing. Train-test splitting helps evaluate how well the model performs on unseen data.

21. Logistic Regression

Logistic Regression is a supervised Machine Learning algorithm used for binary classification problems. In this project, Logistic Regression predicts whether a patient has heart disease or not. Advantages: • Simple and fast • High accuracy • Easy interpretation • Efficient for binary classification

22. K-Nearest Neighbors (KNN)

KNN is a supervised learning algorithm based on nearest neighbors. Working: • Calculates distance between points. • Finds nearest K neighbors. • Uses majority voting for prediction. Advantages: • Easy implementation • Good classification performance • Useful for pattern recognition

23. Model Evaluation Metrics

The models were evaluated using the following metrics: Accuracy: Measures percentage of correct predictions. Precision: Measures correctness of positive predictions. Recall: Measures ability to identify positive cases. F1-Score: Balance between precision and recall. These metrics help compare model performance effectively.

24. Confusion Matrix

Confusion Matrix is used to analyze classification performance. It shows: • True Positives • True Negatives • False Positives • False Negatives Confusion Matrix helps understand prediction correctness and model errors.

25. ROC Curve

ROC Curve (Receiver Operating Characteristic Curve) measures classification performance. The curve compares: • True Positive Rate • False Positive Rate Higher AUC score indicates better model performance.

26. Final Prediction System

A new patient entry was provided to the trained models. The prediction system: • Takes patient details as input. • Applies feature scaling. • Uses trained Logistic Regression and KNN models. • Predicts whether the patient has heart disease. The system successfully generated predictions for new data.

27. Advantages

• Fast disease prediction • Early diagnosis support • Reduces manual analysis • Improves healthcare assistance • Cost-effective system • Easy implementation • Supports medical decision-making

28. Limitations

• Uses synthetic dataset • Real diagnosis still requires doctors • Accuracy depends on data quality • Limited algorithms used • Real-world medical data is more complex

29. Future Scope

The project can be improved by: • Using real hospital datasets • Adding Deep Learning models • Developing a web application • Creating mobile healthcare applications • Real-time prediction systems • Cloud integration • Deployment in hospitals

30. Conclusion

The Heart Disease Prediction System successfully demonstrated the implementation of Machine Learning in healthcare applications. The project included: • Dataset generation • Data cleaning • Visualization • Feature scaling • Logistic Regression • KNN • Model evaluation • Final prediction The models achieved good accuracy and successfully predicted heart disease using patient clinical data. This project improved understanding of Machine Learning, preprocessing, visualization, and classification techniques.

Evaluation Metrics Table

Metric	Description
Accuracy	Measures overall correct predictions
Precision	Measures correctness of positive predictions
Recall	Measures ability to detect positive cases
F1-Score	Balance between precision and recall